1260

MID-TERM EXAMINATION (2026 - 27)

SUBJECT- SCIENCE (Code – 086)

TIME 3 Hours | **CLASS - X** | **M.M. : 80**

General instructions :-

Read the following instructions carefully and follow them :-

(i) This question paper contains 39 questions. All questions are compulsory.

(ii) Question paper is divided into THREE sections – A, B and C.

Section A : Biology (30 Marks)

Section B : Chemistry (25 Marks)

Section C : Physics (25 Marks)

(iii) The question paper has MCQs, VSAs, SAs, LAs and CS/BQs. Marks are given against each question.

(iv) There are case-based questions (CBQs) with three sub-questions and are of 4 marks each.

(v) Divide your answer sheet into three sections as per question paper – Section A (Biology), Section B (Chemistry) and Section C (Physics). It is compulsory to answer each question in its respective section. Do not mix answers of one section into the other section.

(vi) Instructions are given with each section and question, wherever necessary.

(vii) There is no overall choice in the question paper. However, an internal choice has been provided in a few questions. Only one of the choices in such questions must be attempted.

SECTION – A (BIOLOGY)

Q-1 Sensory nerve of a reflex arc carries information from the receptor cells to the :- [1]

(a) Spinal cord (b)Brain

(c) Muscles of the effector organ (d) Bones of the receptor organ

Q-2 The breakdown of pyruvate to give carbon dioxide , water and energy take place in :- [1]

(a) Cytoplasm (b) Mitochondria

(c) Nucleus (d) Chloroplast

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Q-3 Select the mis-match pair :

(a)	Adrenaline	:	Pituitary gland
(b)	Testosterone	:	Testes
(c)	Estrogen	:	Ovary
(d)	Thyroxin	:	Thyroid gland

Q-4 Binary fission is the means of asexual reproduction in which of the following ?

(a) Amoeba (b) Paramecium

(c) Hydra (d) All of these

Q-5 The structural and functional unit of excretion in human beings is:

(a) Glomerulus (b) Nephron

(c) Lymph nodes (d) Malpighian tubules

Q-6 Bile juice is secreted by : [1]

a) Stomach b) Pancreas

c) Small intestine d) Liver

Q-7 In human being, the fertilization occurs in the :- [1]

(a) Uterus (b) Ovaries

(c) Fallopian tube (d) vagina

Directions : Question numbers 8 and 9are Assertion (A) and Reason (R) based questions. Answer these questions by selecting the appropriate option given below:-

A. Both A and R are true, and R is the correct explanation of A.

B. Both A and R are true, and R is not the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Q-8 Assertion (A) - In angiosperms , the ovule develops into seed after fertilization . [1]

Reason (R) – Fertilization is not essential for the development of fruit

Q-9 Assertion (A) : Plant hormones like auxins help in the bending of shoots towards light. [1]

Reason (R) : Auxins accumulate on the side of the shoot facing the light, causing faster cell elongation there.

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Q-10 Draw a neat diagram to show germination of pollen on the female reproductive part of the flower. Name and label only the following parts :- [2]

(a) The part that receives the pollen grain.

(b) The structure that carries the male germ cell to reach the female germ cell.



Q-11 Name the hormone secreted by pancreas and write its function . [2]

OR

(a) Define reflex arc.

(b) Trace the sequence of events which occur in our body when a bright light is focussed on your eyes.

Q-12 Differentiate between sensory and motor neurons. (Any 2 points) [2]

Q-13 Draw and explain how the nerve cells help in transmission of impulses? [3]

Q-14 (i) Name the mode of reproduction of the following organisms and state the important feature of each mode : [3]

a. Hydra

b. Rhizopus

(ii) We can develop new plants from the leaves of Bryophyllum. Comment.

Q-15 Female Reproductive System :- [4]

When a girl is born, the ovaries already contain thousands of immature eggs. On reaching puberty, some of these start maturing. One egg is produced every month by one of the ovaries. The egg is carried from the ovary to the womb through a thin oviduct or Fallopian tube. The two oviducts unite into an elastic bag-like structure known as the uterus. The uterus opens into the vagina through the cervix.

Answer the following questions :-

a) What is fertilisation?

b) Where does fertilization occur?

c) What is placenta?

d) What are the different parts of the female reproductive system?

OR

d) What happens when the egg is not fertilized?

Q-16 Answer the following questions :- [5

a) Name a vein which carries oxygenated blood . - Pulmonary vein.

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b) Name the blood vessel that brings deoxygenated blood to human heart. pulmonary vein

c) Why is left ventricle of heart thicker than right ventricle ?

d) Explain the process of double circulation with the help of flow chart.

OR

Q-16 a) List the three events that occurs during the process of photosynthesis.

b) Explain the two functions of stomata and draw open stomatal pore.

SECTION – B (CHEMISTRY)

Q-17 A metal ribbon 'X' burns in oxygen with a dazzling white flame forming a white ash 'Y'. The correct description of X, Y and the type of reaction is :- [1]

(a) X = Ca ; Y = CaO ; Type of reaction = Decomposition

(b) X = Mg ; Y = MgO ; Type of reaction = Combination

(c) X = Al ; Y = Al_2O_3 ; Type of reaction = Thermal decomposition

(d) X=Zn ; Y = ZnO ; Type of reaction = Endothermic

Q-18 The electron dot structure of chlorine molecule is : [1]

(a) 

(b) 

(c) 

(d) 

Q-19 C_3H_8 belongs to the homologous series of : [1]

(a) Alkynes (b) Alkenes

(c) Alkanes (d) Cycloalkanes

Q-20 How many double bonds are there in a saturated hydrocarbon? [1]

(a) One (b) Two

(c) Three (d) Zero

Q-21 Oils on treating with hydrogen in the presence of palladium or nickel catalyst form fats. This is an example of : [1]

(a) Addition reaction (b) Substitution reaction

(c) Displacement reaction (d) Oxidation reaction

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Q-22 An alloy is called an amalgam if it contains one of the following metals as a constituent :- [1]

(a) Aluminium

(b) Zinc

(c) Tin

(d) Mercury

Q-23 Aqua regia is a mixture of : [1]

(a) 1 part concentrated hydrochloric acid (HCl) and 3 parts concentrated nitric acid (HNO_3)

(b) 3 parts concentrated hydrochloric acid (HCl) and 1 part concentrated nitric acid (HNO_3)

(c) Equal parts of concentrated hydrochloric acid (HCl) and concentrated nitric acid (HNO_3)

(d) Concentrated hydrochloric acid (HCl) and concentrated nitric acid (HNO_3) mixed in any ratio

Directions : Question numbers 8 and 9 are Assertion (A) and Reason (R) based questions.. Answer these questions by selecting the appropriate option given below:

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C. A is true but R is false.

D. A is false but R is true.

Q-24 Assertion (A) : Reaction of Quicklime with water is an exothermic reaction. [1]

Reason (R) : Quicklime reacts vigorously with water releasing a large amount of heat.

Q-25 A student is performing an experiment to study the properties of acetic acid. Answer the following questions :- [2]

(i) Name the substance he must add to acetic acid to produce carbon dioxide.

(ii) Give the relevant chemical equation for the reaction.

(iii) How would he test CO2 gas in the laboratory?

Q-26 (i) Write two observations when lead nitrate is heated in a test tube? [3]

(ii) Name the type of reaction.

(iii) Write a balanced chemical equation to represent the above reaction.

Q-27 Consider the following salts : [3]

(i) YCl (ii) NH_4X (iii) ZCO_3

(a) What would be the pH of the salt solution if, in YCl, Y is sodium? Give reason for your answer.

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(b) If in salt NH4X, X is nitrate, then its solution will give what colour with universal indicator? Why?

(c) What would be the change in colour in blue litmus solution if ZCO_3 is added to it and Z is potassium?

Q-28 When electricity is passed through an aqueous solution of sodium chloride (called brine), it decomposes to form sodium hydroxide. This process is called the chlor-alkali process because the products formed are chlorine (chlor-) and sodium hydroxide (alkali). [4]

Based on the above information, answer the following questions :-

a) Write the chemical equation involved in this process.

b) What are the substances formed at the anode and cathode in the chlor-alkali process?

c) What are the uses of chlorine?

d) Where is the sodium hydroxide solution formed?

e) What are the uses of sodium hydroxide?

Q-29 (a) Show on a diagram the transfer of electrons between the atoms in the formation of MgO. [2]

(b) Why are aqueous solutions of ionic compounds able to conduct electricity? [2]

(c) Name the solvent in which ionic compounds are generally soluble. [1]

SECTION ~~B~~ C (PHYSICS)

Q-30 In which of the following is a concave mirror used ? [1]

~~(a)~~ A solar cooker

~~(b)~~ A rear view mirror in the vehicles.

(c) A safety mirror in shopping malls

(d) In viewing full size image of distant tall buildings.

OR

Q-30 Power of the lens is -40, its focal length is :

a) 4m. b) –40m

c) –0.25m . d) –25m

Q-31 A current-carrying circular loop produces a magnetic field at its center. If the radius of the loop is doubled and the current is halved, the magnetic field at the center becomes : [1]

~~a)~~ One-fourth of original

b) One-half of original

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c) One-eighth of original

d) Same as original

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C. A is true but R is false.

D. A is false but R is true.

[1]

Q-32 Assertion (A) : A concave lens is used to correct myopia.

Reason (R) : Myopia occurs when the image of distant objects is formed in front of the retina.

Q-33 (a) A person is suffering from both myopia and hypermetropia. What kind of lenses can correct this defect? [2]

(b) A person needs a lens of power + 3 D for correcting his near vision and – 3 D for correcting his distant vision. Calculate the focal lengths of the lenses required to correct these defects.

Q-34 (a) Write the two factors on which strength of magnetic field produced by a current carrying conductor depends. [2]

(b) State Flemings left hand rule

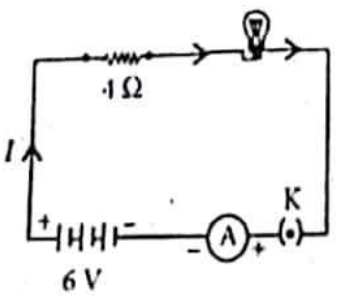
Q-35 (i) Why is an alternating current (A. C.) considered to be advantageous over direct current (D. C.) for the long-distance transmission of electric power? [3]

(ii) How is the type of current used in household supply different from the one given by a battery of dry cells?

(iii) How does an electric fuse prevent the electric circuit and the appliances from possible damage due to short-circuiting or overloading?

OR

Q-35 An electric lamp of resistance 20 Ω and a conductor of resistance 4 Ω. are connected to a 6 V battery as shown in the circuit. Calculate.



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(a) the total resistance of the circuit
(b) the current through the circuit,
(c) the potential difference across the (i) electric lamp and (ii) conductor, and

Q-36 (a) List the factors on which the resistance of a conductor in the shape of a wire depends. [3]

(b) Why are metals good conductors of electricity while glass is a bad conductor of electricity? Give reasons.

(c) Why are alloys commonly used in electrical heating devices? Give reason.

Q-37 (a) Draw magnetic field lines around a bar magnet. [3]

(b) Write all the properties of magnetic field lines.

Q-38 Read the passage carefully and answer the following questions :- [4]

Riya was sitting in the last row of her classroom and found it difficult to read the words written on the blackboard. After an eye check-up, the doctor advised her to wear spectacles with a concave lens. During a science exhibition, she also observed that a rainbow is formed due to the dispersion of sunlight by tiny water droplets. She learned that the human eye has a crystalline lens whose focal length changes with the help of ciliary muscles to focus objects at different distances. The retina contains rod and cone cells that help us see in dim light and in colour.

a) Which defect of vision was Riya suffering from?

b) How can it be corrected ?

c) Write the causes of the defect .

d) Draw the diagram for its correction.

Q-39 The image of a candle flame placed at a distance of 30cm from a spherical lens is formed on a screen placed on the other side of the lens at a distance of 60 cm from the optical centre of the lens. Identify the type of lens and calculate its focal length. If the height of the flame is 3cm, find the height of the image. [5]

OR

Q-39 a) State law of refraction.

b) What is understood as lateral displacement of light? Illustrate this with the help of a diagram. List two factors on which the lateral displacement in a particular substance depends.

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